The Invariance Principle — problem 1.11

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In Fig. 1.5, you may switch the signs of all numbers of a row, column, or a parallel to one of the diagonals. In particular, you may switch the sign of each corner square. Prove that at least one -1 will remain in the table.

1	1	1	1
1	1	1	1
1	1	1	1
1	−1	1	1

Fig. 1.5